D3.1 Reproduction (Human)

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d3-1-human.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Reproduction in flowering plants (D3.1.8–12) is in Plant Systems. Image credits are on the last slide of the slides for this subtopic.

Why reproduce sexually?

D3.1.1 How sexual and asexual reproduction differ

pp. 693–694

	Asexual	Sexual
Parents	One	Usually two
Gametes	None	Haploid gametes fuse
Offspring	Genetically identical	New combinations of genes
Relative advantage	Suits individuals already adapted to an existing environment	Provides the variation needed to adapt to a changed environment
Examples	Yeast, hydra, aphids in summer	Most animals, including humans

Some organisms do both. Aphids reproduce asexually when conditions are stable in spring and summer, and sexually in autumn.

D3.1.2 What meiosis and the fusion of gametes do in a sexual life cycle

pp. 694–695

Meiosis halves the chromosome number to make haploid gametes. **Fertilization**, the fusion of gametes, restores the diploid number. Without meiosis, the number would double every generation. These two events also create variation. Meiosis **breaks up** parental combinations of alleles, and random fusion of gametes **produces new** combinations, unique to the new individual.

Watch out: 23 and 46 apply to humans only. Define a haploid cell as having one set of chromosomes.

D3.1.3 How male and female sexes differ in sexual reproduction

p. 695

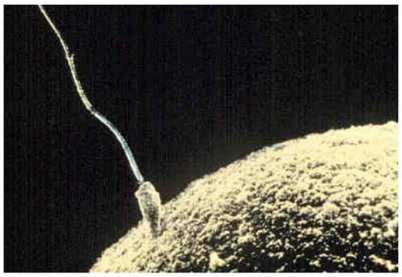
The prime difference is that the **male gamete travels to the female gamete**. So it is small, with few food reserves. Few reach the egg, so very large numbers must be made.

The female gamete does not travel. It is larger, with food reserves for the early embryo, and far fewer are made.

In many animal species, this difference leads to different reproductive strategies: males compete and can mate many times, while females invest more in each offspring and may be more selective. These are broad patterns across species, with many exceptions.

Male and female gametes differ

D3.1.3



A sperm at the surface of an egg. Public domain.

	Male gamete	Female gamete
Movement	Travels	Does not travel
Size	Very small	Much larger
Food reserves	Few	Many
Number	Very many	Few

Key idea
The male gamete **travels** to the female one. So it is small and light, many are made because few arrive, and the egg carries the food.

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Human reproductive systems

D3.1.4 The parts of the human male and female reproductive tracts

pp. 695–697

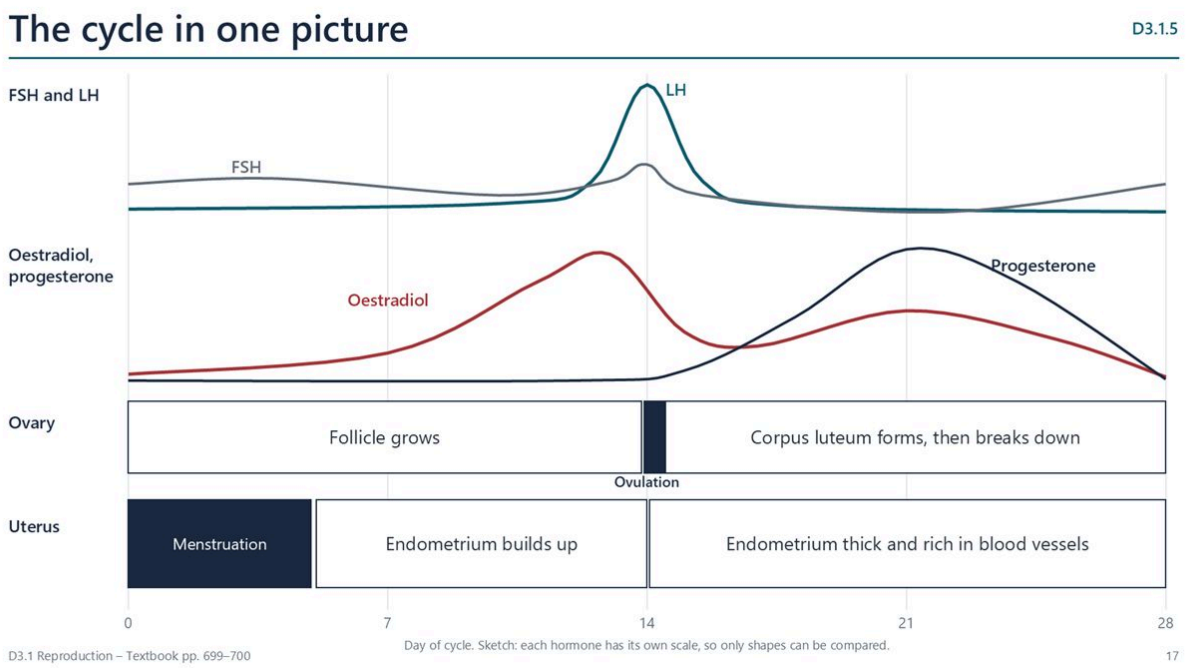
Male-typical	Function	Female-typical	Function
Testes	Make sperm and testosterone	Ovaries	Make eggs, oestradiol and progesterone
Scrotum	Keeps testes cooler than the body	Oviducts	Carry the egg; site of fertilization
Epididymis	Stores sperm	Uterus	Muscular wall; embryo develops here
Sperm duct	Carries sperm to the urethra	Endometrium	Lining rebuilt each cycle
Seminal vesicles and prostate	Add fluid to make semen	Cervix	Connects uterus and vagina
Urethra and penis	Carry semen out during ejaculation	Vagina	Muscular tube to the outside

Exam note: you need to be able to draw and annotate both systems. Side views show positions best. Check your connections: the sperm duct must join the urethra, and each oviduct must open into the space inside the uterus.

D3.1.5 What happens in the ovarian and uterine cycles, and how hormones control them

pp. 698–700

The menstrual cycle lasts about 28 days and combines an **ovarian cycle**, preparing and releasing an egg, and a **uterine cycle**, building up and shedding the endometrium.



1. Menstruation marks day 1. The pituitary gland secretes **FSH**, which stimulates follicles to develop.
2. A follicle secretes **oestradiol**, which rebuilds the endometrium. It also increases FSH receptors in the follicle, so more oestradiol is made: **positive feedback**.
3. High oestradiol inhibits FSH, so no more follicles develop: **negative feedback**.
4. High oestradiol also stimulates a surge of **LH**, which triggers ovulation at about day 14.
5. LH converts the empty follicle into a **corpus luteum**, which secretes **progesterone** and some oestradiol.
6. Progesterone maintains the endometrium and inhibits FSH and LH: **negative feedback**.
7. Low LH and FSH let the corpus luteum break down. Progesterone and oestradiol fall, and the endometrium is shed.

If an embryo implants, it secretes a hormone that keeps the corpus luteum going, so progesterone stays high.

Watch out: FSH stimulates follicle growth, while LH triggers ovulation. When progesterone falls and FSH rises, that is the removal of negative feedback and not positive feedback.

Fertilization and IVF

D3.1.6 How fertilization happens in humans

p. 701

Fertilization is internal, usually in the upper part of the oviduct.

1. The cell membranes of the sperm and egg fuse.
2. The sperm nucleus enters the egg. The tail and mitochondria of the sperm are destroyed, so all of the new cell's mitochondria come from the egg.
3. The nuclear membranes of both nuclei break down.
4. The condensed chromosomes from both nuclei take part in a joint mitosis, producing two diploid nuclei.

Watch out: avoid saying that the two nuclei fuse into one.

D3.1.7 How hormones are used in in vitro fertilization (IVF)

pp. 701–702

IVF is fertilization outside the body. Normal secretion of hormones is first suspended, so that doctors control the timing.

- **Down-regulation:** a drug, often a nasal spray, stops pituitary and ovarian hormones for about two weeks.
- **Superovulation:** high doses of FSH make many follicles develop together.
- **Maturation:** an injection of hCG brings the eggs to maturity. They are collected about 36 hours later.
- **Fertilization and transfer:** eggs are mixed with sperm. The embryos are incubated, and one or more are placed in the uterus.
- **Progesterone** is given to maintain the endometrium.

Not every treatment leads to a birth, and transferring several embryos raises the risk of a multiple pregnancy.

How hormones are used in IVF

D3.1.7

Stage	What is given	Why
Down-regulation	A drug for about two weeks	Stops the natural hormones, so timing is controlled
Superovulation	High doses of FSH, about ten days	Many follicles develop at once
Maturation	An hCG injection	Eggs mature; collected about 36 hours later
Fertilization	No hormone	Eggs mixed with sperm in a dish
Implantation	Progesterone	Keeps the endometrium thick

One or two embryos are then placed in the uterus.

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